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DEBBIE: 427,022 Galaxy Candidates with No Prior Identification in the DESI Legacy Imaging Surveys DR10

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ABSTRACT

I present DEBBIE, 427,022 low surface brightness galaxy candidates selected from the DESI Legacy Imaging Surveys DR10 source catalogue, none of which the galaxy catalogues I searched, NED, SIMBAD or DESI DR1 identify as a galaxy. Candidates are chosen by cuts on Tractor size, surface brightness and data quality, and screened against 1.1×10^8 catalogued galaxies. Visual inspection of stratified samples finds a galaxy at the catalogued position for 76–89% of entries. Boosted trees and a bidirectional recurrent network trained on radial light profiles score every entry with usable imaging. No entry is removed on its score, but the 57% with a mean score above 0.2 are 95–97% galaxies. Within the surface-brightness window the cuts keep 56–89% of known galaxies with effective radii of 3–15″ and almost none above. All redshifts are photometric. The designations, DEBBIE NNNNNN, are registered with the IAU.

1. INTRODUCTION

Imaging surveys catalogue every source they detect, and those catalogues contain galaxies that no galaxy catalogue lists. DEBBIE selects them from the DR10 sweep catalogues of the DESI Legacy Imaging Surveys (Dey et al. 2019), which hold a Tractor model fit (Lang et al. 2016) for every detection. It adds no detections: every entry is a Tractor source, and the claim is only that none of the catalogues and services below identifies it as a galaxy. The name stands for DR10 Ensemble of Boosted trees and Bidirectional recurrent networks for Identifying Extragalactic sources. A full paper is in preparation.

2. SELECTION

I keep every source in the DR10 south sweeps with `TYPE` `EXP`, `DEV`, `SER` or `REX`, which excludes point sources; $r_{\rm eff}$ = `SHAPE_R` $\geq 3''$; mean surface brightness within $r_{\rm eff}$, $\mu_{\rm eff} = g + 2.5\log_{10}(2\pi r_{\rm eff}^2)$, between 24.0 and 26.5 mag arcsec^{-2}, with g from `FLUX_G` and not corrected for extinction; `NOBS_G` and `NOBS_R` ≥ 2; `FLUX_G`, `FLUX_R` and `FLUX_Z` > 0; and `MASKBITS` $= 0$. Fainter than 26.5, Tractor increasingly fits large radii to noise. The main footprint is $|b| \geq 30°$, about 12,900 deg^2, and a flagged extension covers $12° < |b| < 30°$ with the same cuts.

3. SCREENING

A candidate is removed if (1) a galaxy in a local index lies within $\max(2r_{\rm eff}, 10'')$; the index holds 110,190,191 objects from HyperLEDA (Makarov et al. 2014), SDSS DR12 photometric galaxies (Alam et al. 2015), GLADE+ (Dálya et al. 2022), the 2MASS Extended Source Catalog (Jarrett et al. 2000), 6dFGS (Jones et al. 2009) and SMUDGes (Zaritsky et al. 2023); (2) it lies within half the D_{25} diameter of a HyperLEDA galaxy, which removes fragments of large galaxies; (3) NED or SIMBAD (Wenger et al. 2000) lists an object of a galaxy, quasar or AGN type within $\max(5'', r_{\rm eff})$, and a failed NED query also removes it; (4) DESI DR1 (DESI Collaboration 2026) has a spectrum with `ZWARN` $= 0$ within $2''$, whatever its class; or (5) three or more other candidates lie within $1.5'$, which happens on defective imaging. DES DR2 sources (Abbott et al. 2021) do not count against a candidate, since that catalogue lists sources, not identifications. Rows within $\max(6'', r_{\rm eff})$ of each other are merged, keeping the largest.

The catalogue has three parts that share one number sequence (Table 1). The original part, at $|b| \geq 30°$, and 78,234 of the extension entries were screened with fixed radii in place of rule (3), removing any candidate with a NED galaxy within $30''$, any SIMBAD object within $15''$, or any index object, DES DR2 included, within $45''$. The recovered part holds the candidates at $|b| \geq 30°$ that those wider radii removed and that pass rules (1)–(5); the rest of the extension was recovered in the same way.

4. SCORES

For each entry with usable imaging I sample the g, r and z images in 24 annuli and 32 position angles about the catalogued position and divide all three bands by one amplitude, so that profile shape and colour survive and brightness does not. Gradient-boosted trees and a bidirectional recurrent network score these profiles, and the catalogue gives both scores and their mean. Trained on the DeepShadows labels (Tanoglidis et al. 2021), low surface brightness galaxies against artefacts, the trees reach AUC 0.981 on that benchmark. Against my visual verdicts the mean score separates galaxies from the rest with AUC 0.85 in the original part, 0.87 in the recovered part and 0.83 in the extension. In the original part's sample, dropping entries with a mean score below 0.06 keeps 95% of the galaxies and removes 28% of the rest. A second pair, trained on HyperLEDA galaxies against stars and faint Tractor fits, reached AUC 0.41 and 0.61 on the two samples and is not released. No entry is removed on its score.

5. PURITY

I judged by eye 999 objects from a sample of the original part and 2,002 from a sample of the other two parts, on Legacy Surveys colour images. Both samples are stratified over ($r_{\rm eff}$, $\mu_{\rm eff}$) cells and weighted by the entries each object represents. A galaxy lies at the catalogued position for 89.1% of the original part, 76.1% of the recovered part and 81.9% of the extension, leaving out objects I could not decide; Table 1 also gives each figure with those counted as non-galaxies. Purity falls close to catalogued galaxies, and the fixed radii kept the original part at least 45″ from them; in the second sample it is 60% within 20″ of one and 84% beyond 30″. Entries with a mean score above 0.2, 70% of the original part and 48% of the others, are 95.4–96.5% galaxies in each part. One person made every verdict.

6. COMPLETENESS AND CAVEATS

Measured on catalogued galaxies in the footprint, the cuts alone keep 56–89% of those with $r_{\rm eff} = 3$–15″ and $\mu_{\rm eff}$ in the window, but 1 of 34 above 15″, because Tractor splits large galaxies into fragments that fail the cuts. DEBBIE cannot constrain large diffuse galaxies. The screen makes the catalogue uneven: between right ascensions 150° and 200° it holds 2.1 entries per deg^2 inside the SDSS footprint ($4° < \delta < 20°$), against 24.8 just south of it ($-10° < \delta < -4°$). In the original part 63% of the redshifts come from Duncan (2022) and the rest from a five-band model I trained on those, which suits population statistics only. The median is 0.29 and only 2,591 entries (1.6%) lie at $z < 0.05$, so most are distant galaxies rather than nearby diffuse dwarfs.

7. DESIGNATIONS AND DATA

Designations take the form DEBBIE NNNNNN and are registered with the IAU through the CDS Dictionary of Nomenclature. Numbers run from 3 to 453450 with gaps, because a number is assigned once and never reused. Positions are Tractor positions on the Gaia frame. A flux-weighted centroid on the DR10 images lies within 0.40″ of the catalogued position for half the entries, and more than 2.5″ away, or could not be measured, for 15%. The catalogue, with each entry's Tractor photometry and shape, part, scores and centroid offset, is at Zenodo (doi:10.5281/zenodo.23659669).

Table 1. The three parts. Purity: weighted share of inspected objects with a galaxy at the catalogued position, undecided objects left out (95% interval) or counted as non-galaxies. AUC: mean score of the DeepShadows-trained pair against the same verdicts. Centroid: share of entries whose centroid lies within 1″.

Part	$\lvert b\rvert$	Entries	Purity (%)	Undecided as no (%)	AUC	Centroid (%)
Original	$\geq 30°$	161,533	89.1 (86.4–91.6)	83.7	0.85	81.9
Recovered	$\geq 30°$	141,405	76.1 (73.0–78.9)	76.1	0.87	66.5
Extension	12–30°	124,084	81.9 (79.2–84.3)	81.8	0.83	65.5
All		427,022				72.0

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data.desi.lbl.gov/doc/acknowledgments/, https://www.darkenergysurvey.org/the-des-project/data-access/ and https://www.sdss3.org/collaboration/boiler-plate.php. This publication makes use of data products from the Two Micron All Sky Survey (Skrutskie et al. 2006), which is a joint project of the University of Massachusetts and the Infrared Processing and Analysis Center/California Institute of Technology, funded by the National Aeronautics and Space Administration and the National Science Foundation. The NASA/IPAC Extragalactic Database is funded by NASA and operated by the California Institute of Technology. SIMBAD and VizieR are operated at CDS, Strasbourg. I acknowledge the use of the HyperLEDA database (http://leda.univ-lyon1.fr). I thank Clay, Raina, Lu, Trishy, Angie, Gigi, Tuna, Vesna, Joe, Ambre, and all my other friends who have supported me in my eccentricity.

Software: XGBoost, PyTorch, NumPy, SciPy, Astropy

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